

LAPORAN PRAKTIKUM – MERGE QUICK SORT

Laporan ini disusun untuk memenuhi Tugas Ke Enam Praktikum Algoritma & Pemrograman

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PROGRAM STUDI TEKNIK INFORMATIKA

FAKULTAS TEKNIK

UNIVERSITAS PANCASILA

2025/2026

Soal Quick Sort_16 (nama file: PASD6-16.cpp):

- **Source Code**

16. Source code dibawah ini apabila di RUN, salah. Silahkan perbaiki programnya
(nama file : PASD6-16.ccp):

```
#include<iostream>
using namespace std;

int a[50];

void Merge(int,int,int);

void MergeSort(int low,int high){
    int mid;
    if(low<high){
        mid=(low+high)/2;
        MergeSort(low,mid);
        MergeSort(mid+1,high);

        Merge(low,mid,high); }}

void Merge(int low,int mid,int high){
    int h,i,j,b[50],k;
    h=low;
    i=low;
    j=mid+1;
    while((h<=mid)&&(j<=high)){
        if(a[h]<=a[j]){
            b[i]=a[h];
            h++;}
        else{
            b[i]=a[j];
            j++;}
        i++;}
```

```

    if(h>mid){
        for(k=j;k<=high;k++){
            b[i]=a[k];
            i++;}}
    else{
        for(k=h;k<=mid;k++){
            b[i]=a[k];
            i++;}}
    for(k=low;k<=high;k++){
        a[k]=b[k];}}

int main(){
    int num,i;
    cout<<"~~~~~"<<endl;
    cout<<" PROGRAM PENGURUTAN MERGE "<<endl;
    cout<<"~~~~~"<<endl;
    cout<<endl;
    cout<<"Masukkan jumlah bilangan yang diinginkan: ";
    cin>>num;
    cout<<"-----"<<endl;
    cout<<"Masukkan    sejumlah    "<<num<<"    data    yang    akan
diurutkan"<<endl;
    for(i=1;i<=num;i++){
        cout<<"Bilangan ke-"<<i<<" : ";
        cin>>a[i];}
    MergeSort(1,num);
    cout<<endl;
    cout<<"Hasil Akhir Pengurutan : "<<endl;
    cout<<endl;
    for(i=1;i<=num;i++){
        cout<<a[i]<<" ";}
    cout<<endl;
    cin.get();}

```

- SS Program

```
#include<iostream>
using namespace std;

int a[50];

void Merge(int,int,int);

void MergeSort(int low,int high){
    int mid;
    if(low<high){
        mid=(low+high)/2;
        MergeSort(low,mid);
        MergeSort(mid+1,high);

        Merge(low,mid,high); }}

void Merge(int low,int mid,int high){
    int h,i,j,b[50],k;
    h=low;
    i=low;
    j=mid+1;
    while((h<=mid) && (j<=high)){
        if(a[h]<=a[j]){
            b[i]=a[h];
            h++;}
        else{
            b[i]=a[j];
            j++;}
        i++;}
    if(h>mid){
        for(k=j;k<=high;k++){
            b[i]=a[k];
            i++;}}
    else{
        for(k=h;k<=mid;k++){
            b[i]=a[k];
            i++;}}
```

```

        for (k=low;k<=high;k++) {
            a[k]=b[k];}

int main() {
    int num,i;
    cout<<"~~~~~"<<endl;
    cout<<" PROGRAM PENGURUTAN MERGE "<<endl;
    cout<<"~~~~~"<<endl;
    cout<<endl;
    cout<<"Masukkan jumlah bilangan yang diinginkan: ";
    cin>>num;
    cout<<"-----"<<endl;
    cout<<"Masukkan sejumlah "<<num<<" data yang akan diurutkan"<<endl;
    for (i=1;i<=num;i++) {
        cout<<"Bilangan ke- "<<i<<" : ";
        cin>>a[i];}
    MergeSort(1,num);
    cout<<endl;
    cout<<"Hasil Akhir Pengurutan : "<<endl;
    cout<<endl;
    for (i=1;i<=num;i++) {
        cout<<a[i]<<" ";}
    cout<<endl;
    cin.get();}

```

- **SS Running Program**

```

D:\ASD SMT2\ASD-A-25-26\PRAK6_Merge Quick Sort_Asprak01_45252110035_Khairul Adam Efendi>PASD6-16
PROGRAM PENGURUTAN MERGE
~~~~~
Masukkan jumlah bilangan yang diinginkan: 10
-----
Masukkan sejumlah 10 data yang akan diurutkan
Bilangan ke-1 : 45
Bilangan ke-2 : 25
Bilangan ke-3 : 21
Bilangan ke-4 : 0
Bilangan ke-5 : 0
Bilangan ke-6 : 35
Bilangan ke-7 : 11
Bilangan ke-8 : 66
Bilangan ke-9 : 55
Bilangan ke-10 : 21

Hasil Akhir Pengurutan :

0 0 11 21 21 25 35 45 55 66

```

Soal Merge Sort_25 (nama file: PASD6-25.cpp):

- **Source Code**

25. Berdasarkan contoh no2, buat dengan menggunakan class, dengan inputan dan melakukan pengurutan secara Descending (nama file : PASD6-25.ccp):

```
#include<iostream>
using namespace std;

class MergeSort{
private:
    int arr[100];
    int n;

public:

    void inputData(){
        cout<<"Masukkan jumlah data : ";
        cin>>n;
        cout<<"Masukkan "<<n<<" data : "<<endl;
        for(int i=0;i<n;i++){
            cout<<"Data ke-"<<i+1<<" : ";
            cin>>arr[i];}}

    void merge(int Kiri,int mid,int Kanan){
        int n1=mid-Kiri+1;
        int n2=Kanan-mid;
        int L[100],R[100];
        for(int i=0;i<n1;i++){
            L[i]=arr[Kiri+i];}
        for(int j=0;j<n2;j++){
            R[j]=arr[mid+1+j];}
        int i=0;
        int j=0;
        int k=Kiri;
```

```
while(i<n1&& j<n2){
```

```
    if(L[i]>=R[j]){
```

```
        arr[k]=L[i];
```

```
        i++;}
```

```
    else{
```

```
        arr[k]=R[j];
```

```
        j++;}
```

```
    k++;}
```

```
while(i<n1){
```

```
    arr[k]=L[i];
```

```
    i++;
```

```
    k++;}
```

```
while(j<n2){
```

```
    arr[k]=R[j];
```

```
    j++;
```

```
    k++;}}
```

```
void mergeSort(int Kiri,int Kanan){
```

```
    if(Kiri<Kanan){
```

```
        int mid=Kiri+(Kanan-Kiri)/2;
```

```
        mergeSort(Kiri,mid);
```

```
        mergeSort(mid+1,Kanan);
```

```
        merge(Kiri,mid,Kanan);}}
```

```
void cetakData(){
```

```
    for(int i=0;i<n;i++){
```

```
        cout<<arr[i]<<" ";}
```

```
    cout<<endl;}
```

```
void proses(){
```

```
    cout<<endl;
```

```
    cout<<"Data Sebelum Disort"<<endl;
```

```
    cout<<"~~~~~"<<endl;
```

```
        cetakData();
        mergeSort(0,n-1);
        cout<<endl;
        cout<<"Data Setelah Disort Descending"<<endl;
        cout<<"~~~~~"<<endl;
        cetakData();}

};

int main(){
    MergeSort M;
    M.inputData();
    M.proses();
    return 0;}
```


- SS Program

```
#include<iostream>
using namespace std;

class MergeSort{
private:
    int arr[100];
    int n;

public:

    void inputData() {
        cout<<"Masukkan jumlah data : ";
        cin>>n;
        cout<<"Masukkan " <<n<<" data : "<<endl;
        for(int i=0;i<n;i++){
            cout<<"Data ke-"<<i+1<<" : ";
            cin>>arr[i];}}

    void merge(int Kiri,int mid,int Kanan) {
        int n1=mid-Kiri+1;
        int n2=Kanan-mid;
        int L[100],R[100];
        for(int i=0;i<n1;i++){
            L[i]=arr[Kiri+i];}
        for(int j=0;j<n2;j++){
            R[j]=arr[mid+1+j];}
        int i=0;
        int j=0;
        int k=Kiri;
        while(i<n1&& j<n2) {
            if(L[i]>=R[j]){
                arr[k]=L[i];
                i++;}
            else{
                arr[k]=R[j];
                j++;}
```

```

        k++;}
    while(i<n1){
        arr[k]=L[i];
        i++;
        k++;}
    while(j<n2){
        arr[k]=R[j];
        j++;
        k++;}}

void mergeSort(int Kiri,int Kanan){
    if(Kiri<Kanan){
        int mid=Kiri+(Kanan-Kiri)/2;
        mergeSort(Kiri,mid);
        mergeSort(mid+1,Kanan);
        merge(Kiri,mid,Kanan);}}

void cetakData(){
    for(int i=0;i<n;i++){
        cout<<arr[i]<<" ";}
    cout<<endl;}

void proses(){
    cout<<endl;
    cout<<"Data Sebelum Disort"<<endl;
    cout<<"~~~~~"<<endl;
    cetakData();
    mergeSort(0,n-1);
    cout<<endl;
    cout<<"Data Setelah Disort Descending"<<endl;
    cout<<"~~~~~"<<endl;
    cetakData();}

};

int main(){
    MergeSort M;
    M.inputData();
    M.proses();
    return 0;}

```

- **SS Running Program**

```
D:\ASD SMT2\ASD-A-25-26\PRAK6_Merge Quick Sort_Asprak01_45252110035_Khairul Adam Efendi>PASD6-25
Masukkan jumlah data : 10
Masukkan 10 data :
Data ke-1 : 45
Data ke-2 : 25
Data ke-3 : 21
Data ke-4 : 0
Data ke-5 : 0
Data ke-6 : 35
Data ke-7 : 11
Data ke-8 : 66
Data ke-9 : 55
Data ke-10 : 21

Data Sebelum Disort
45 25 21 0 0 35 11 66 55 21

Data Setelah Disort Descending
66 55 45 35 25 21 21 11 0 0
```